Md Shahanur Islam Shagor
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I am writing to apply for the Computer Vision & AI Engineer position on the N3XT Interceptor C-UAS team at Quantum-Systems GmbH. The opportunity to work on perception systems for autonomous interceptor platforms operating under extreme latency and motion constraints aligns directly with the type of engineering problems I pursue in my research and project work.

Quantum-Systems is building autonomous sUAS platforms that require robust, low-latency perception in defense and security contexts. This focus on real-world deployment of computer vision and AI on embedded hardware is what draws me to this role.

Through research and development work, I have built practical experience with computer vision pipelines optimized for embedded execution. I developed a Wireless Vision-Aid for the Blind (WVAB) using YOLOv8n for real-time object detection, running entirely offline on a Raspberry Pi with a custom UDP streaming protocol for low-latency communication. This project required optimizing inference for constrained hardware, working with camera pipelines, and converting image-space detections into actionable directional cues.

I also designed OmniGuard V2X, a privacy-preserving blockchain framework for autonomous vehicle networks, deployed on NVIDIA Jetson Nano. This involved integrating camera-based detection with sensor fusion, implementing secure communication protocols, and building data pipelines for training and validation. The work required combining classical computer vision approaches with machine learning models, benchmarking different architectures, and deploying inference on embedded systems.

In addition to my technical experience, I work effectively across multidisciplinary teams and communicate clearly with both technical and non-technical stakeholders. I am fluent in Bengali and English, have working knowledge of Russian and Hindi, and enjoy collaborating in international and cross-cultural environments.

I have attached my CV and cover letter for your review. I would welcome the opportunity to discuss how my hands-on experience with perception pipelines, embedded AI, and autonomous systems can contribute to the N3XT Interceptor platform.

Sincerely,

Md Shahanur Islam Shagor
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